

Beyond the Flight Recorder: A Cryptographic Biometric Framework for Bridging Individual and Organizational Aviation Risk

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*Preprint submitted to **Safety Science** (Q1, Elsevier)*

Not yet peer-reviewed — Istanbul, 2026

Abstract

Every pilot error is an organizational failure (Reason, 2000; Dekker, 2007). Aviation safety has achieved remarkable maturity through the systematic development of reactive and proactive paradigms over the past half century. Yet the mechanical telemetry provided by flight data recorders (FDR) and cockpit voice recorders (CVR) cannot capture the cognitive and physiological state of the pilot at the moment of decision-making, creating a structural diagnostic gap in accident investigations. This study presents a two-phase cryptographic data processing architecture developed within the Complementary Diagnostic Safety Approach (CDSA) framework to address this gap. In the first phase, HRV, SpO₂, and GSR data from smart pilot watches are encrypted at the device level using AES-256-GCM; the master key is then split into three shares via Shamir Secret Sharing and distributed to independent institutional parties. The 2-of-3 threshold scheme ensures that individual biometric data can only be accessed following an accident report with simultaneous authorization from at least two parties. In the second phase, macro biometric data anonymized through differential privacy mechanisms ($\epsilon = 1.0$, k -anonymity $k = 5$) are integrated into Safety Management Systems (SMS) to enable proactive risk analysis. Validated at prototype level in Python, the architecture is compliant with the privacy-by-design principle under GDPR Article 25 and with Just Culture philosophy. All reported findings are derived from a literature-calibrated synthetic dataset ($n = 1,179$); real operational validation constitutes the primary agenda of future research. The study also proposes a Biometric Flight Data Declaration within ICAO Annex 13.

Keywords: Aviation Safety; Complementary Diagnostic Safety Approach (CDSA); Cryptographic Privacy; Shamir Secret Sharing; Differential Privacy; Wearable Biometric Sensors

1. Introduction

Aviation engineering has achieved an extraordinary level of mechanical reliability over the course of its history. According to data from the International Civil Aviation Organization (ICAO), the accident rate per million flights in commercial aviation has decreased approximately fortyfold since the early 1970s (International Civil Aviation Organization, 2022). Yet behind this remarkable achievement in mechanical safety, a structural gap persists that has not been fully resolved: the inability to explain, at a diagnostic level, the human factors underlying aviation accidents.

A review of the aviation safety literature reveals that existing approaches have coalesced around two fundamental paradigms. The reactive paradigm seeks to prevent the recurrence of similar events by conducting systematic analyses following accidents and

incidents (Gerede, 2018). The proactive paradigm aims to identify risks before they materialize, working to elevate operational safety through tools such as Safety Management Systems (SMS) and the Human Factors Analysis and Classification System (HFACS) (Shappell and Wiegmann, 2000). While the contributions of these two paradigms to aviation safety are undeniable, both are structurally incapable of objectively recording the pilot’s cognitive and physiological state at the moment of an accident.

The traditional accident investigation tools—the FDR and CVR—capture the aircraft’s mechanical responses and cockpit conversations. Yet what the pilot was experiencing at those critical moments, the level of oxygen saturation, or how heart rate variability was evolving under acute stress, remains largely in the dark (Dehais et al., 2020). This situation creates fertile ground for the proliferation of superficial causal attributions such as “pilot error” in accident reports—attributions that simultaneously conceal genuine root causes and render systemic failures invisible (Reason, 2000).

Recent advances in wearable sensor technologies carry considerable potential for closing this diagnostic gap. Smart pilot watches and biometric sensors can measure parameters such as HRV, SpO₂, and GSR in real time; when fused with traditional flight telemetry, these data offer the possibility of reconstructing the physiological state at the moment of an accident with mathematical precision (Cantekin, 2025). However, the obstacles to realizing this technological potential are less technical than ethical and legal. A pilot’s biometric data—heart rate, stress level, sleep quality—are classified as special category personal data under the European Union’s General Data Protection Regulation (GDPR), and questions of ownership and access rights over such data remain unresolved critical issues (Erdoğan, 2020).

This study proposes the Complementary Diagnostic Safety Approach (CDSA)—a framework that addresses these ethical and technical challenges simultaneously—and presents in detail the cryptographic infrastructure underpinning this approach. CDSA is positioned as a third safety layer that goes beyond reactive and proactive paradigms. Its core innovation lies in a two-phase data processing architecture based on AES-256-GCM encryption and Shamir Secret Sharing protocol. Through this architecture, individual biometric data are protected by cryptographic guarantees while anonymized macro data contribute to proactive risk management.

The original contributions of this study can be summarized across four dimensions: (1) the reconceptualization of aviation safety within a three-layer paradigm framework encompassing reactive, proactive, and diagnostic approaches; (2) the design of a cryptographic architecture integrating AES-256-GCM and Shamir Secret Sharing with Just Culture principles; (3) the adaptation of differential privacy mechanisms to aviation safety management; and (4) the operationalization of the CDSA ecosystem through publicly accessible prototype applications on the cdsa.app platform. The remainder of the paper is organized as follows: Section 2 reviews the relevant literature; Section 3 presents the

CDSA framework and cryptographic architecture; Section 4 reports prototype implementation and validation findings; Section 5 discusses ethical and regulatory dimensions; and Section 6 evaluates the study’s contributions and future research directions.

2. Related Work

The following five axes of literature are reviewed: the evolution of safety paradigms in aviation, wearable biometric sensor applications, cryptographic privacy architectures, data privacy and Just Culture principles, and the identification of the gap that this study addresses.

2.1. The Evolution of Paradigms in Aviation Safety

A field’s understanding of safety is directly linked to how the accidents within that field are interpreted. A review of the aviation safety literature reveals that the discipline’s history of more than a century has been built on a reactive foundation. Early accident investigations relied on linear causality models in which events were attributed to a single cause—typically a mechanical failure or human error. Reason (2000)’s Swiss Cheese model fundamentally transformed this understanding, demonstrating that accidents arise from the simultaneous alignment of gaps across multiple defensive layers. This model established systemic thinking in aviation and formed the theoretical basis for proactive safety management.

The operational counterpart of proactive safety approaches is ICAO’s Safety Management System (SMS) framework (International Civil Aviation Organization, 2020). SMS seeks to systematically integrate hazard identification, risk assessment, and safety assurance cycles into institutional operations. The HFACS model developed by Shappell and Wiegmann (2000) classifies human factor-related accidents within a four-level hierarchical structure, providing accident investigators with a powerful analytical tool. Despite these advances, Gerede (2018) emphasizes that existing proactive tools remain structurally limited in their diagnostic capacity precisely because they cannot access the physiological data layer at the moment of an accident.

2.2. Wearable Biometric Sensors and Aviation Applications

The potential of wearable technologies in the aviation context has attracted growing interest over the past decade. Heart rate variability (HRV) serves as a direct indicator of autonomic nervous system activity, objectively reflecting the pilot’s cognitive workload and stress level (Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, 1996). Dehais et al. (2020), in a study conducted from a neuroergonomics perspective, demonstrated that HRV reduction reliably predicts prefrontal cortex overload and consequent loss of situational awareness. Peripheral oxygen saturation (SpO₂) functions as a critical biomarker for the early de-

tection of covert hypoxia risk, particularly under high-altitude conditions. [Denison and Ledwith \(1972\)](#) demonstrated that the pilot’s reasoning capacity begins to deteriorate well before SpO₂ drops below 91 %.

On the integration of biometric data with flight safety, [Ratti and Truscon \(2021\)](#) showed that sensor fusion algorithms, when combined with FDR telemetry, can reconstruct the causal chain at the moment of an accident with mathematical accuracy. [Cantekin \(2025\)](#) comprehensively examined the diagnostic potential of pilot watches evolving from the mechanical era to AI-assisted systems and laid out the first systematic framework of the CDSA concept.

2.3. Cryptographic Privacy Architectures

The most critical legal and ethical dimension of biometric data processing is the provision of data privacy guarantees at the design level. [Cavoukian \(2009\)](#)’s Privacy by Design principle advocates that personal data protection mechanisms be conceived as an integral part of the architecture rather than added to the system after the fact. This principle has been elevated to the status of a legal obligation under Article 25 of the GDPR ([European Parliament, 2016](#)).

In the field of threshold cryptography, [Shamir \(1979\)](#) developed a protocol enabling a secret value to be split into n shares such that any $k \geq 2$ shares can reconstruct it via Lagrange interpolation. Differential privacy, formally defined by [Dwork and Roth \(2014\)](#), renders individual re-identification statistically impossible by adding controlled random noise to a dataset. The application of cryptographic access control and threshold schemes to aviation data management finds very limited coverage in the literature; the NTSB’s ([2023](#)) projection report explicitly articulates the need for a regulatory framework in this area.

The relationship between regulatory complexity and cryptographic data protection architecture is not unique to the aviation sector. In healthcare, HIPAA and GDPR mandate that patient data be protected through encryption, access control, and audit trail mechanisms; compliance must be grounded in direct technical safeguards rather than institutional trust alone ([Conduah et al., 2025](#)). In the financial sector, SOX, MiFID II, and Basel III frameworks require comprehensive data governance infrastructures, and the compliance burden simultaneously creates structural barriers to market entry for new technological actors ([McNulty et al., 2023](#)). In defense and intelligence, ITAR and FedRAMP High certifications demand compliance processes spanning more than a decade. A common pattern emerges: regulatory cryptographic obligations strengthen the market position of compliant platforms; privacy-enhancing technologies are increasingly regarded not merely as compliance tools but as prerequisites for value creation in the data economy ([National Science and Technology Council, 2022](#)). CDSA represents a sector-specific application of this cross-industry data protection paradigm.

2.4. Data Privacy, Just Culture, and Pilot Rights

The point at which biometric monitoring systems intersect with pilot rights constitutes the most contested axis in the literature. The Just Culture principle conceptualized by Dekker (2007) holds that errors should be reported not to be punished but to identify vulnerabilities in the system. The Air Line Pilots Association (2022) documented in its technical white paper that the absence of this guarantee would fundamentally undermine pilot trust and produce a counterproductive effect on safety culture. Ustaömer and Şengür (2020) demonstrated that the institutional trust environment is the single most critical factor determining safety reporting behavior. Erdinç (2020) evaluated the legal requirements for processing biometric data under the special category personal data framework through the lenses of proportionality and informed consent.

2.5. The Gap in the Literature and the Position of This Study

Two fundamental observations emerge from the body of literature summarized above. First, studies addressing the integration of wearable biometric sensors into aviation diagnostic processes discuss the technical opportunity framework but fall short of proposing an end-to-end cryptographic architecture to secure this integration. Second, the data privacy and Just Culture literature evaluates the ethical framework broadly but cannot systematically demonstrate how these principles translate into engineering design. This study aims to close these gaps through the integrated application of AES-256-GCM real-time encryption, Shamir 2-of-3 threshold cryptography, and Laplace differential privacy.

3. CDSA Framework and Cryptographic Architecture

This section presents the conceptual foundation of CDSA and its two-phase cryptographic architecture in detail. Phase 1 addresses individual diagnostic access control; Phase 2 covers anonymized macro-level data analysis for proactive risk management.

3.1. Conceptual Framework: The Third Safety Paradigm

A system’s diagnostic capacity is determined not only by the effectiveness of its preventive mechanisms but also by the extent to which it can illuminate the causal chain underlying events. Viewed from this perspective, the current reactive and proactive paradigms in aviation safety share a common blind spot: both are structurally incapable of objectively recording the pilot’s physiological and cognitive state at the moment of an accident.

The CDSA proposed in this study is positioned as a third safety layer targeting precisely this blind spot. The core thesis of CDSA can be stated as follows: when biometric data obtained from wearable sensors are fused with traditional flight telemetry

at millisecond precision, accident investigations can move beyond superficial attributions such as “pilot error” and physiological root causes—covert hypoxia, acute stress, or startle effect—can be established with mathematical certainty. CDSA seeks to resolve the tension between individual privacy and diagnostic access not through legal regulation but through the architecture of cryptography itself.

3.2. Phase 1: Cryptographic Diagnostic Data Architecture

Phase 1 comprises a five-layer hierarchical architecture designed to make individual pilot biometric data accessible only following an accident or serious incident report and subject to simultaneous multi-party authorization.

3.2.1. System Overview

Phase 1 of CDSA is built on a five-layer hierarchical architecture: (1) real-time biometric measurement and encryption, (2) splitting of the master key into shares via threshold cryptography, (3) distribution of key shares to independent institutional parties, (4) unlocking of the locked box following an accident report and with threshold authorization, and (5) decryption and sensor fusion with FDR/CVR data. Figure 1 summarizes the architecture and data flow across these five layers.

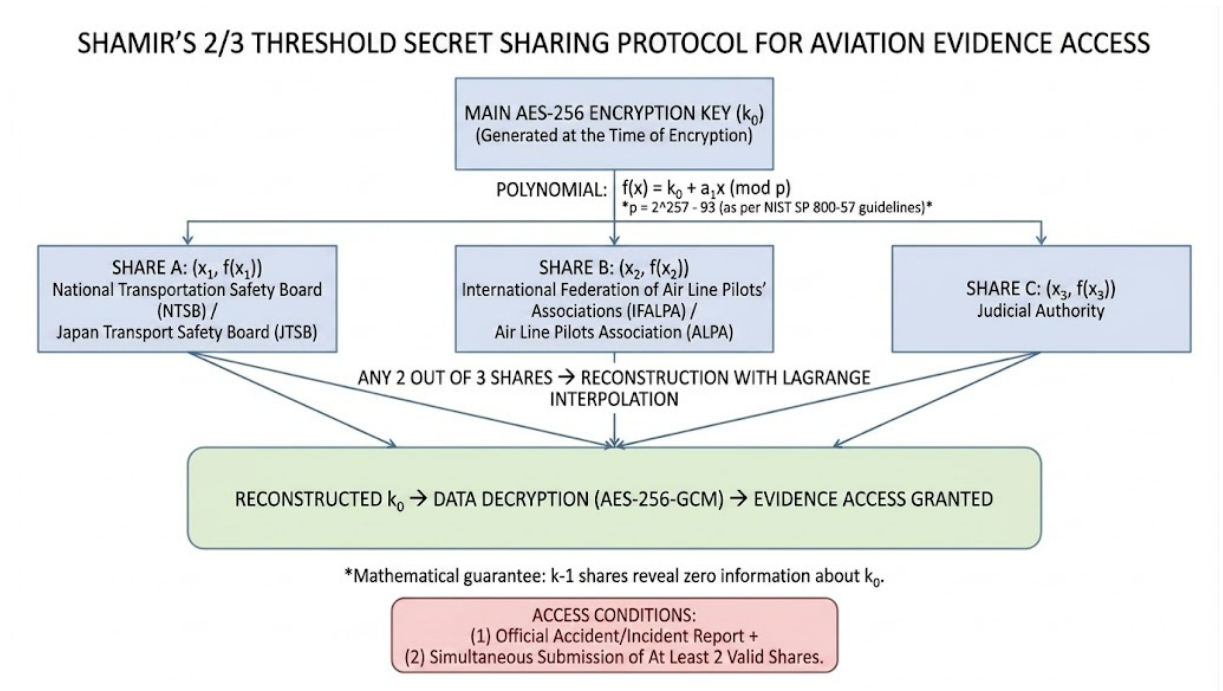


Figure 1: CDSA Two-Phase Cryptographic Architecture. Phase 1 (left): AES-256-GCM encryption, Shamir 2/3 threshold key distribution, and locked-box access control. Phase 2 (right): Differential privacy anonymization and SMS integration. Both phases share the same hardware infrastructure while maintaining cryptographic separation of access.

3.2.2. Data Acquisition and Real-Time Encryption

The most effective form of data security is considered to be ensuring that sensitive information is never stored in plaintext. Accordingly, the design principle adopted is that HRV, SpO₂, and GSR values continuously acquired from the smart pilot watch at millisecond precision are never written to device memory in raw form. Each measurement is encrypted in real time at the hardware level using the AES-256-GCM algorithm (NIST SP 800-38D). The 96-bit nonce value used in encryption is generated independently for each record by a cryptographically secure pseudorandom number generator (CSPRNG), ensuring that identical data never produce the same ciphertext. Each record additionally carries an Additional Authentication Data (AAD) field composed of the flight identifier, timestamp, and box identifier; this field is cryptographically bound to the GCM authentication tag.

3.2.3. Shamir Secret Sharing and 2-of-3 Threshold Cryptography

To eliminate the security vulnerability created by a single center’s absolute control over the master key, the AES-256 master key (k_0) is split into $n = 3$ shares using the Shamir Secret Sharing protocol (Shamir, 1979). The mathematical foundation of the system is expressed as:

$$f(x) = k_0 + a_1x \pmod{p}, \quad p = 2^{257} - 93 \quad (1)$$

$$S_i = (x_i, f(x_i)) \longrightarrow \text{Share } i \text{ transmitted to Party } i \quad (2)$$

Here p is a 257-bit prime compliant with NIST SP 800-57 standards. The threshold parameter $k = 2$ means that any two of the three shares suffice to reconstruct the master key exactly via Lagrange interpolation. A single share held by one party is mathematically meaningless and cannot yield the original key even with unlimited computational power. Table 1 summarizes the cryptographic parameters of the CDSA architecture.

Table 1: CDSA Cryptographic Architecture Parameter Summary.

Parameter	Value / Description
Symmetric encryption	AES-256-GCM (NIST SP 800-38D)
Nonce length	96-bit (CSPRNG, independent per record)
Secret sharing protocol	Shamir Secret Sharing (Shamir, 1979)
Finite field	$p = 2^{257} - 93$ (257-bit prime, NIST SP 800-57)
Threshold parameters	$k = 2, n = 3$ (2-of-3 threshold scheme)
AAD field	Flight ID + Timestamp + Box ID
Integrity control	GCM authentication tag

Within the parameter framework summarized in Table 1, Figure 2 illustrates the

protocol flow through which Shamir Secret Sharing splits the master key into three independent shares and reconstructs it when the threshold number of shares is combined.

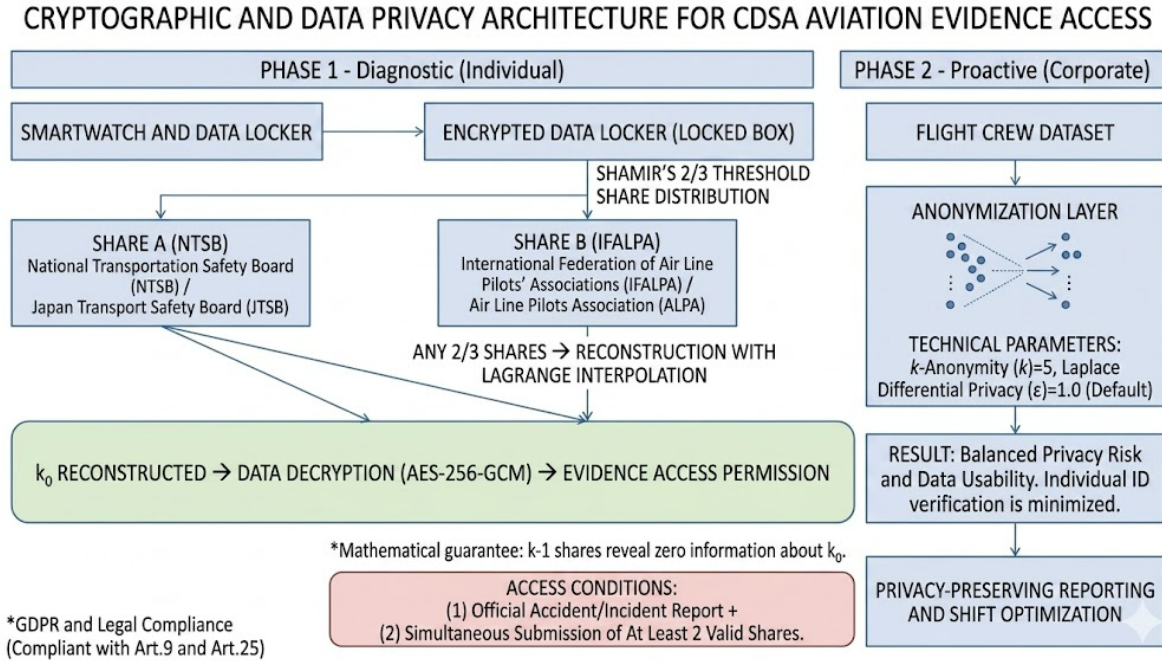


Figure 2: Shamir Secret Sharing: 2-of-3 Threshold Cryptography Protocol. The AES-256 master key k_0 is split into $n = 3$ shares using a polynomial over a finite field ($p = 2^{257} - 93$). Any 2 shares allow reconstruction via Lagrange interpolation; a single share alone reveals zero information.

3.2.4. Independent Key Holders and Institutional Balance

The effectiveness of cryptographic systems is measured not only by their mathematical strength but also by the institutional independence of their key holders. Accordingly, the three key shares are distributed to three independent institutional parties whose interests are structurally opposed:

- **Share A — Accident Investigation Authority (NTSB/JTSB):** Submits the key share upon completion of accident reporting. Its legal status, independent of commercial pressures, constitutes the primary safeguard.
- **Share B — Pilot Union (IFALPA/ALPA):** Represents pilot consent in its capacity as the guardian of labor rights. Retains veto power over the use of data in disciplinary proceedings.
- **Share C — Judicial Authority:** Activated solely upon court order and upon presentation of legal justification.

This tripartite structure renders it mathematically impossible for any single party to unlock the system alone, while simultaneously aligning with Just Culture principles (Dekker, 2007).

3.2.5. Locked Box Access Control and Decryption Protocol

During routine flight operations, encrypted biometric data accumulate in hardware units referred to as the locked box. After the flight concludes and the box is sealed, the master key is cryptographically deleted from device memory. At this stage, even if the airline or aircraft manufacturer physically possesses the encrypted data, they cannot access its contents. The decryption mechanism can only be triggered by the simultaneous fulfillment of two conditions: the filing of an accident or serious incident report, and the simultaneous submission of key shares by at least two parties. The decrypted HRV, SpO₂, and GSR time series are synchronized with FDR telemetry at millisecond precision and presented to the accident investigation board.

3.3. Phase 2: Macro Data Analysis with Differential Privacy

Phase 2 shares the same biometric hardware infrastructure as Phase 1 but is fully decoupled from it through cryptographic access control. In this phase, anonymized macro data containing no individual identity are processed for proactive risk management.

3.3.1. Anonymization Architecture

While Phase 1 of CDSA serves a diagnostic function, Phase 2 leverages the same biometric data infrastructure for proactive risk management purposes. Data stripped of individual pilot identity pass through two complementary privacy mechanisms. The first mechanism is based on the k -anonymity ($k = 5$) principle (Sweeney, 2002). The second mechanism is Laplace differential privacy ($\epsilon = 1.0$) (Dwork and Roth, 2014):

$$\mathcal{M}(x) = x + \text{Lap}\left(\frac{\Delta f}{\epsilon}\right), \quad \Delta f = \text{sensitivity}, \quad \epsilon = \text{privacy budget} \quad (3)$$

The sensitivity values applied are $\Delta f = 5.0 \text{ ms}$ for HRV, $\Delta f = 1.0\%$ for SpO₂, and $\Delta f = 2.0 \mu\text{S}$ for GSR.

3.3.2. SMS Integration and Proactive Risk Outputs

Anonymized macro data are integrated with SMS to produce two types of operational output: (1) evidence-based justification for shift optimization in route and shift combinations where high SRS values are concentrated; and (2) support for ICAO Annex 6 FDP limitations through the finding that night shifts produce statistically significantly higher risk profiles across all routes. In neither output does individual pilot data enter the processing pipeline.

3.4. Comparative Summary of the Architecture

Figure 3 below presents a visual comparative summary of the two phases across data type, encryption mechanism, access condition, authorized users, primary goal, and GDPR basis.

CDSA DATA FLOW: PHASE 1 (DIAGNOSTIC) vs PHASE 2 (PROACTIVE)		
DIMENSION	PHASE 1 — DIAGNOSTIC	PHASE 2 — PROACTIVE
Data Type	Individual — full identifiable	Anonymized — no individual record
Privacy Mechanism	AES-256-GCM + Shamir 2/3	k-Anonymity (k=5) + Laplace Differential Privacy ($\epsilon=1.0$)
Access Trigger	Accident / Serious Incident Report	Continuous — real-time SMS feed
Authorized Users	Accident Investigation Board + Relevant Unions + Judicial Authorities	Airline SMS Managers
Primary Goal	Root cause reconstruction	Proactive operational risk management
GDPR Basis	Art. 9 (sensitive) + Art. 25 (PbD)	Art. 6 (legitimate interest)
Output	Biometric evidence timeline + FDR fusion	Shift optimization · FDP recommendations

** Both phases run on the same biometric hardware — cryptographic access control provides strict separation*

Figure 3: CDSA Phase 1 vs Phase 2: Comparative Data Flow. Key distinctions across data type, privacy mechanism, access trigger, authorized users, primary goal, GDPR basis, and operational output. Both phases share the same biometric hardware infrastructure.

4. Prototype Implementation and Validation

This section validates the functionality of the CDSA architecture at two complementary levels: a Python-based cryptographic prototype implementation and publicly accessible user interfaces deployed on the cdsa.app platform.

4.1. Validation Strategy and Prototype Scope

The scientific value of a theoretical framework depends in large part on the extent to which its practical feasibility can be concretely demonstrated. The former establishes mathematical correctness; the latter demonstrates operational feasibility.

A synthetic covert hypoxia scenario formed the basis of the prototype development process: $SpO_2 = 91\%$, $HRV = 33.2\text{ms}$, $GSR = 13.96\ \mu\text{S}$, $SRS = 0.457$ (above the critical threshold of 0.40). These parameters are consistent with the literature-calibrated synthetic dataset (seed = 42).

4.2. Cryptographic Prototype — Python Implementation

The cryptographic prototype was developed and validated at two levels: the implementation environment and components are described in Section 4.2.1, while the Phase 1 validation findings are reported in Section 4.2.2.

4.2.1. Implementation Environment and Components

The cryptographic prototype was developed in Python 3.10 using the `cryptography` library for encryption operations. The prototype consists of three mod-

ules: (1) data generation and real-time encryption, (2) key management via Shamir Secret Sharing, and (3) decryption and integrity verification via Lagrange interpolation. The complete code is presented in Appendix C and shared publicly in accordance with the reproducibility principle.

4.2.2. Phase 1 Validation Findings

In the encryption module, six biometric measurement records from the hypoxia scenario defined above were encrypted in real time using AES-256-GCM. Independent 96-bit nonce values generated for each record provided mathematical protection against timing attacks. Successful binding of AAD fields to GCM authentication tags was confirmed.

In the Shamir Secret Sharing module, the master key k_0 was split into $n = 3$ shares over the finite field $p = 2^{257} - 93$, with each share transmitted to the corresponding institutional party. It was verified that using only two of the three shares—NTSB and IFALPA, without the judicial authority’s share—the master key was fully reconstructed via Lagrange interpolation. The key integrity verification test returned a positive result; single-party access attempts were automatically rejected.

The decryption output reconstructed the covert hypoxia scenario with mathematical precision: SpO_2 had dropped to 91%, HRV had declined to 33.2ms, and GSR had risen to $13.96 \mu\text{S}$ at the moment of critical event onset—findings that establish the existence of a physiological root cause chain beyond the attribution of “pilot error.” Figure 4 presents the PilotSense AUTH interface alongside the decrypted biometric evidence timeline for the TC-XXX scenario.

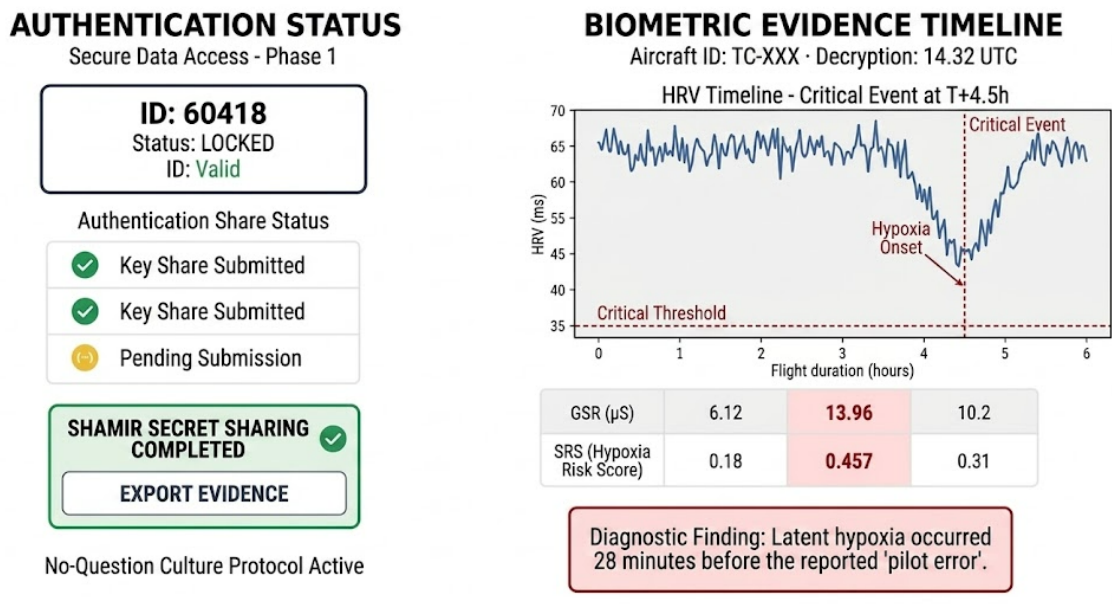


Figure 4: PilotSense AUTH Interface: Phase 1 Diagnostic Access Control. Left: encrypted locked-box with NTSB and IFALPA shares submitted (2/3 threshold reached). Right: decrypted biometric evidence timeline revealing covert hypoxia onset 28 minutes before the reported event (synthetic scenario, TC-XXX).

4.3. Prototype Applications — Overview

The CDSA ecosystem comprises five prototype applications, each targeting a distinct operational layer. Table 2 summarizes the key metrics obtained for each application.

Table 2: Summary of CDSA Prototype Applications.

Application	Layer	Key Metric
PilotSense AUTH	Diagnostic — Phase 1	2/3 multi-party signature, AES-256-GCM, GDPR Art. 25
PilotSense OPS	Proactive — Phase 2	247 pilots, SMS integration, anonymous fleet monitoring
PilotO2	Proactive — Individual	95% model accuracy, 92% hypoxia detection, 127 ms
PilotGuard	Proactive — Individual	87% pre-flight risk, 93% stress detection
PilotReflect	Post-Flight (EFB)	8-min debriefing, ISO 9241-11, Privacy by Design

All applications are publicly accessible on the [cdsa.app](#) platform. It must be emphasized that all five applications remain at the prototype stage and FAA/EASA certification processes have not yet been completed.

4.3.1. PilotSense AUTH — Diagnostic Layer Interface

PilotSense AUTH operationalizes Phase 1 of CDSA at the user interface level. In the TC-XXX simulation scenario, a 14.7MB encrypted biometric data file was successfully decrypted through the simultaneous submission of NTSB and IFALPA shares, demonstrating the practical functionality of the 2/3 threshold scheme. The GDPR Article 25 compliance layer remained active throughout.

4.3.2. PilotSense OPS — Proactive Fleet Monitoring Interface

PilotSense OPS represents Phase 2 of CDSA, targeting airline SMS managers and operations centers. In the simulation environment, anonymized biometric data from 247 active pilots were successfully processed; eight warnings and two critical conditions were automatically detected. The SRS values of critical records exceeded the threshold of 0.40.

4.3.3. PilotO2 — Covert Hypoxia Early Warning System

The PilotO2 application is built on a three-layer LSTM neural network architecture consisting of 128, 64, and 32 processing units. The system achieved 95% model accuracy, 92% hypoxia detection rate, and 127 ms response time. Validation under real operational conditions constitutes a priority for future research.

4.3.4. PilotGuard — Pre-Flight Readiness Assessment

PilotGuard processes HRV, instantaneous heart rate, stress index, and sleep quality data in real time to generate an integrated pilot flight readiness score. Simulation tests yielded 87% pre-flight risk prediction and 93% stress detection accuracy, with a user experience rating of 4.6/5.0.

4.3.5. PilotReflect — EFB Integrated Biometric Debrief System

PilotReflect delivers post-flight personal biometric assessment within the Type B EFB ecosystem. In line with [Cavoukian’s \(2009\)](#) Privacy by Design principles, all biometric data are stored exclusively on the pilot’s personal device encrypted with AES-256. An average debriefing duration of eight minutes and an FAA AC 120-76E compliant interface design were achieved in accordance with ISO 9241-11 usability criteria.

4.4. Integrated Assessment of Prototype Findings

Viewed collectively, the prototype validation findings demonstrate that the CDSA ecosystem can be successfully translated from a theoretical framework into a functional platform. Several limitations must be carefully underscored. All five applications remain at the prototype stage; FAA/EASA certification processes have not yet been completed. The cryptographic validation was conducted on a synthetic dataset, and sensor behavior under real operational conditions—G-forces, airframe vibration—has not yet been tested.

5. Discussion

This section examines the ethical coherence of the cryptographic architecture, the Panopticon paradox and its design solution, compliance with GDPR and the international regulatory framework, the limitations of synthetic data, comparative positioning in the literature, and practical feasibility.

5.1. Ethical Coherence of the Cryptographic Architecture

For a technological solution to be sustainable, it must satisfy not only engineering criteria but also the value systems of the communities it affects. The most distinctive contribution of the CDSA architecture is considered to lie not in its technical innovation but in its ethical coherence. Existing aviation data systems attempt to manage the tension between privacy and diagnostic access through legal regulation—an approach whose guarantees shift whenever legislation changes or interpretations diverge. CDSA transfers this tension to the mathematical foundations of cryptography, thereby eliminating this fragility.

The alignment of the 2-of-3 threshold scheme with Just Culture philosophy is particularly noteworthy in this context. In the CDSA architecture, the pilot union’s possession of a key share prevents biometric data from being used in disciplinary proceedings not

through a legal guarantee but through a mathematical impossibility. As [Air Line Pilots Association \(2022\)](#) emphasizes, pilot organizations systematically oppose biometric monitoring systems precisely because of their distrust of existing legal safeguards—making the mathematical nature of the guarantee, rather than its legal form, the critical differentiator.

5.2. The Panopticon Paradox and the Design Solution

The most insidious risk created by continuous biometric monitoring is not the direct harm it may cause to the pilot but the behavioral change it may induce. [Foucault’s \(1992\)](#) Panopticon metaphor illustrates this mechanism compellingly: an individual who knows they are under constant surveillance begins to behave as though being watched even in the absence of observation. Transposed to the aviation context, this effect carries the potential to amplify performance anxiety, elevate allostatic load, and paradoxically threaten flight safety ([Zuboff, 2019](#)).

CDSA’s response to this paradox is direct and structural. During routine flight operations, airline management mathematically cannot access individual biometric data. The pilot knows that the system is not monitoring them at the individual level; they are only aware that their anonymized macro data contribute to fleet-wide risk patterns. This design choice is considered to eliminate the Panopticon effect. It must be acknowledged, however, that this guarantee holds only as long as the system functions correctly; a security breach or data leak could fundamentally destabilize this balance.

5.3. Compliance with GDPR and the International Regulatory Framework

The legal positioning of the CDSA architecture simultaneously engages multiple regulatory layers. The individual biometric data processed in Phase 1 are classified as special category personal data under GDPR Article 9 ([European Parliament, 2016](#)). CDSA addresses this requirement by integrating pilot consent into key management through Share B—an approach that functions as both a technical and a legal safeguard. The anonymized macro data in Phase 2 fall outside the scope of GDPR’s definition of personal data following the application of differential privacy mechanisms. At the international level, a Biometric Flight Data Declaration is presented as this study’s concrete policy proposal for updating the ICAO Annex 13 framework ([International Civil Aviation Organization, 2020](#)).

5.4. Limitations of Synthetic Data and Ecological Validity

Addressing the methodological limitations of this study openly and honestly is a fundamental requirement of academic integrity. The biometric data used to validate the cryptographic architecture were derived from a literature-calibrated synthetic dataset ($n = 1,179$, seed = 42). This dataset has established its statistical validity through Shapiro-Wilk subgroup normality testing (13/16 subgroups, $p > 0.05$), Spearman cor-

relation ($\rho = -0.648$, $p < 0.001$), and Mann-Whitney U ($p < 0.001$) tests; however, its ecological validity under real cockpit conditions has not yet been demonstrated. Figure 5 presents the HRV–GSR correlation structure and the route-by-shift SRS risk heatmap derived from this synthetic dataset.

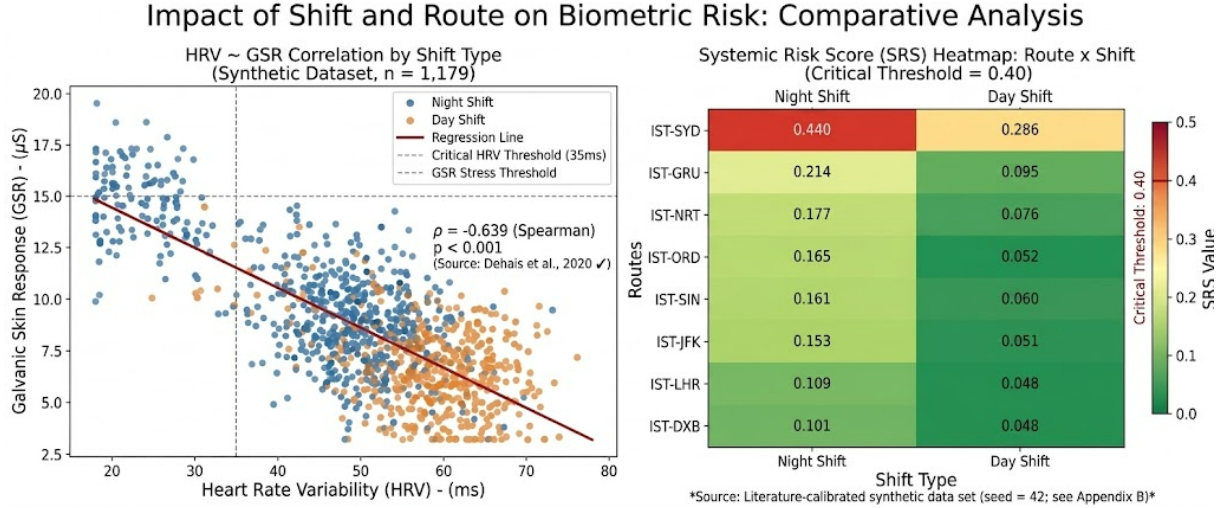


Figure 5: HRV ~ GSR Correlation and SRS Risk Heatmap (Synthetic Dataset, $n = 1,179$, seed = 42). Left: Spearman $\rho = -0.648$, $p < 0.001$, consistent with Dehais et al. (2020). Right: Route $\times$ Shift SRS heatmap. IST-SYD Night (SRS = 0.440) approaches the critical threshold of 0.40. Source: Literature-calibrated synthetic data (Appendix B).

Three ecological validity concerns are identified. First, the extent to which high G-forces and airframe vibration introduce noise into biometric sensor measurements is unknown. Second, variance in individual pilot baseline values is not represented in the synthetic data, rendering it impossible to personalize the SRS critical threshold on a per-pilot basis. Third, the impact of sensor calibration drift on system performance during extended operational use has not yet been modeled. Addressing these limitations constitutes the primary objective of the three-phase validation plan presented in Section 6.3.

5.5. Comparative Positioning: CDSA in the Literature

The CDSA architecture sits at the intersection of three related but independently evolving streams of research: physiological monitoring studies in aviation safety literature, threshold scheme applications in applied cryptography, and differential privacy systems in data privacy literature. Dehais et al. (2020) comprehensively documented the diagnostic potential of biometric monitoring yet did not address cryptographic privacy guarantees. Dwork and Roth (2014) established the theoretical foundations of differential privacy on solid ground but did not discuss operational applications in high-risk domains such as aviation. The NTSB’s (2023) projection report proposed the integration of wearable device data into accident investigations yet was unable to offer any technical framework for how this integration should be supported by cryptographic guarantees. CDSA aims

to close these gaps within a single unified architecture.

Beyond the literature gaps identified above, the cryptographic guarantee approach proposed in this study represents a sector-specific application of data protection paradigms long established in other highly regulated industries. In healthcare, HIPAA and GDPR mandate technically grounded data protection through encryption, access control, and audit trail mechanisms (Conduah et al., 2025). In finance, SOX, MiFID II, and Basel III frameworks impose comprehensive data governance obligations that simultaneously create structural entry barriers for new market entrants (McNulty et al., 2023). Across these sectors, a consistent pattern emerges: regulatory cryptographic obligations strengthen the market position of compliant platforms, and privacy-enhancing technologies are increasingly regarded not merely as compliance tools but as prerequisites for value creation in the data economy (National Science and Technology Council, 2022). CDSA, positioned as the aviation-specific instantiation of this cross-industry paradigm, inherits both the methodological legitimacy and the strategic positioning that this broader pattern affords.

5.6. Practical Feasibility and Operational Barriers

The first barrier is certification cost. The DO-178C Level B compliance process demands significant financial and temporal resources. Smaller and medium-sized airlines may struggle to bear this burden independently, potentially delaying sector-wide adoption. Shared certification pools and industry consortia represent viable pathways for overcoming this challenge.

The second barrier is institutional cultural resistance. Pilot organizations harbor deep and legitimate reservations about biometric monitoring (Air Line Pilots Association, 2022). Although the CDSA 2-of-3 threshold scheme renders disciplinary use of data mathematically impossible, translating technical guarantees into institutional trust requires sustained transparency policies and long-term stakeholder dialogue.

The third barrier is the requirement for multi-party coordination. The legal framework governing the simultaneous participation of the accident investigation authority, pilot union, and judicial authority during international flights has not yet been clarified. Incorporation of a “Personal Biometric Flight Data” standard provision into ICAO Annex 13 represents the most operationally feasible solution to this gap.

The fourth barrier is the absence of real-world data. All cryptographic validation work in this study was conducted on synthetic data. The effects of real cockpit conditions—G-forces, airframe vibration, and inter-pilot baseline variability—on system performance remain unknown. This uncertainty limits the confidence of regulators and pilot organizations in the system, making the three-phase validation roadmap presented in Section 6.3 a strategic imperative.

6. Conclusion

This section summarizes the original contributions of the study across four dimensions, states the methodological limitations explicitly, and identifies three priority directions for future research.

6.1. Assessment of Original Contributions

The true maturity of a safety system is measured not only by its capacity to prevent events but also by its ability to fully comprehend the human reality underlying those events. The CDSA framework proposed in this study has been positioned as a third layer complementing the reactive and proactive paradigms of aviation safety, and an integrated cryptographic architecture has been developed capable of illuminating—with mathematical precision—the root causes of human factors in accident investigations.

The four original contributions of this study can be summarized as follows. The first contribution is conceptual: the systematic introduction of the diagnostic paradigm into the literature alongside the reactive and proactive paradigms, carrying the potential to fundamentally transform how human factors are approached in accident investigations. The second contribution is technical: the integration of AES-256-GCM and Shamir 2-of-3 threshold cryptography with Privacy by Design principles, grounding biometric data privacy in a mathematical foundation independent of legal guarantees. The third contribution is methodological: the integration of differential privacy mechanisms ($\epsilon = 1.0$, k -anonymity $k = 5$) with Safety Management Systems, resolving the tension between individual privacy and collective risk management at the statistical level. The fourth contribution is applied: the operationalization of the CDSA ecosystem through publicly accessible prototype applications on the cdsa.app platform, demonstrating the framework’s feasibility beyond abstract proposal. Furthermore, the cryptographic guarantee approach proposed in this study represents a sector-specific application of data protection paradigms long established in healthcare (HIPAA), finance (SOX/MiFID II), and defense (ITAR/FedRAMP) — positioning CDSA as the aviation instantiation of a cross-industry regulatory data protection pattern (Conduah et al., 2025; McNulty et al., 2023; National Science and Technology Council, 2022).

6.2. Limitations

All cryptographic validation findings in this study were conducted on a literature-calibrated synthetic dataset ($n = 1,179$, seed = 42). Completing DO-178C Level B or higher compliance before CDSA components can be operationally deployed does not appear feasible within the current regulatory framework. The legal framework governing the simultaneous participation of the accident investigation authority, pilot union, and judicial authority in international flights has not yet been clarified. Beyond these structural limitations, it must be explicitly acknowledged that the ecological validity of the synthetic

dataset findings under real cockpit conditions—including the effects of G-forces, airframe vibration, and inter-pilot baseline variability—remains unestablished.

6.3. Future Research Roadmap

Three priority research directions emerge from a joint assessment of this study’s findings and limitations. Real operational data validation is the first priority: the collection of real operational biometric data through voluntary pilot participation and in collaboration with IFALPA/ALPA, and the application of the same regression framework, are planned. This validation process is envisioned in a three-phase structure: (1) establishment of ethical approval and data access protocols, (2) Anderson-Darling distribution fit testing and comparison of SRS patterns with real data, and (3) updating of the calibration matrix to transform CDSA into an operational tool. Initiating the FAA/EASA certification process constitutes the second priority. ICAO Annex 13 integration—through the advancement of the Biometric Flight Data Declaration into the policy process—constitutes the third priority.

6.4. Closing: Toward a Sustainable Balance Between Privacy and Diagnostic Power

The future of aviation safety most likely lies in a better understanding of human beings rather than machines. While aircraft have approached a near-perfect level of mechanical reliability, the human factor remains the most complex and least understood component of the system. The wearable technology ecosystem carries the potential to transform this equation—yet this potential must be realized without compromising pilot rights, undermining safety culture, or eroding institutional trust.

The CDSA framework proposed in this study seeks to resolve the tension between privacy and diagnostic power through the design of technology itself. The system’s success ultimately depends on the simultaneous fulfillment of three conditions: validation of the cryptographic architecture against real operational data, updating of the regulatory framework to accommodate this innovation, and the establishment of long-term dialogue with pilot organizations. To the extent that these three conditions can be met concurrently, CDSA will be capable of substantiating its claim to be the third paradigm of aviation safety.

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